$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\frac{\partial f}{\partial x} \quad \frac{\partial^2 f}{\partial x \partial y} \quad \frac{\partial^2 f}{\partial x^2} \quad \frac{\partial^3 f}{\partial x^2 \partial y} \quad \frac{\partial^3 f}{\partial x \partial y \partial x} \quad \frac{\partial}{\partial t} \quad \frac{\partial^2 u}{\partial x_1^2}$$

$$H_f = \begin{bmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} \end{bmatrix}$$

$$H_f = \begin{bmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_3} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} & \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_3} \\ \dfrac{\partial^2 f(v)}{\partial x_3 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_3 \partial x_2} & \dfrac{\partial^2 f(v)}{\partial x_3^2} \end{bmatrix}$$

$$H_g = \begin{bmatrix} \dfrac{\partial^2 g(v)}{\partial y_1^2} & \dfrac{\partial^2 g(v)}{\partial y_1 \partial y_2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_1 \partial y_n} \\ \dfrac{\partial^2 g(v)}{\partial y_2 \partial y_1} & \dfrac{\partial^2 g(v)}{\partial y_2^2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_2 \partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial^2 g(v)}{\partial y_n \partial y_1} & \dfrac{\partial^2 g(v)}{\partial y_n \partial y_2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_n^2} \end{bmatrix}$$

$$H_f(\mathbf{a}) = \begin{bmatrix} \dfrac{\partial^2 f(\mathbf{a})}{\partial x_1^2} & \dfrac{\partial^2 f(\mathbf{a})}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(\mathbf{a})}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(\mathbf{a})}{\partial x_2^2} \end{bmatrix} \qquad H_f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Proposition 1 (Identity)**.** *For every $n \times n$ matrix A we have $I_n A = A$.*

Exercise 2. Compute $\begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$ for $f(x_1, x_2) = x_1^2 x_2$.

By **Proposition 1**, **Exercise 2** needs no identity matrix. See also **Theorem 4.2** in the course notes.

Added from IMO26 notation

$$\frac{\mathrm{d}f}{\mathrm{d}x} \quad \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} \quad \frac{\mathrm{d}}{\mathrm{d}t}$$

$$\nabla f(v) = \begin{bmatrix} \dfrac{\partial f(v)}{\partial x_1} \\ \dfrac{\partial f(v)}{\partial x_2} \\ \dfrac{\partial f(v)}{\partial x_3} \end{bmatrix} \qquad \nabla g = \begin{bmatrix} \dfrac{\partial g}{\partial y_1} \\ \dfrac{\partial g}{\partial y_2} \\ \vdots \\ \dfrac{\partial g}{\partial y_n} \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n, \quad \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}, \quad \begin{bmatrix} w_1 & w_2 & w_3 & w_4 \end{bmatrix}, \quad \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}, \quad A^T$$

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

$$\|x\|, \quad |x - y|, \quad \left\|\frac{u}{2}\right\|, \quad \langle u, v \rangle = u^T v$$

$$\{1, 2, 3\}, \quad \{x \in S \mid p(x)\}, \quad \left\{ x \in \mathbb{R}^n \;\middle|\; \|x\| \leq \frac{1}{2} \right\}$$

$$f \colon \mathbb{R}^n \to \mathbb{R}, \quad x - y \coloneqq x + (-y)$$

$$x^* = \arg\min_{x \in C} f(x), \qquad \min_{\substack{x \in \mathbb{R}^n \\ Ax \leq b}} f(x) \qquad \max_{x \geq 0} c^T x$$

$$H_f = \begin{bmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{12} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix}, \quad \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{12} & s_{22} & s_{23} \\ s_{13} & s_{23} & s_{33} \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}, \quad \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$